Hardware & System Requirements Overview

Here is a simplified breakdown of the hardware you need to run the OmniRetail AI Suite (unified_app.exe), which handles customer counting, demographics, and staff attendance on Windows.

Hardware Options

- **Basic setup (Budget Edge / Mini PC):** Intel Core i3 (8th Gen+) or Ryzen 3 3000, 8 GB RAM, integrated graphics, 128 GB SATA SSD, Windows 10/11, and 100 Mbps network connection.
- **Recommended setup (Standard Retail Store):** Intel Core i5 (10th Gen+) or Ryzen 5 4000/5000, 16 GB RAM, Intel Iris Xe Graphics (for faster OpenVINO processing), 256–512 GB NVMe SSD, Windows 10/11 Pro, and 1 Gbps Ethernet.
- **High-Performance setup (Busy Outlets):** Intel Core i7 (11th Gen+) or Ryzen 7, 16–32 GB RAM, dedicated NVIDIA GPU (GTX 1650 / RTX 3050) or Intel Arc, 512 GB+ NVMe SSD, and 1 Gbps Ethernet.

App Footprint & System Load

- **Memory Usage:** The app takes roughly 1.5 GB to 2.4 GB of RAM while running all background AI models (footfall counting, demographics, face attendance, and video display).
- **CPU Load:** Expect the app to use about 40% to 70% of a modern 4-to-6 core processor.

Key Recommendations

- **CPU Requirements:** Must support AVX2 instructions (standard on modern Intel and AMD processors) so the AI frameworks can run smoothly.
- **RAM Baseline:** Do not use 4 GB systems. Stick with at least 8 GB for basic performance, though 16 GB is strongly recommended to keep things stable 24/7.
- **Intel Mini PC Advantage:** Using OpenVINO with Intel integrated graphics (like Iris Xe) offloads work from the CPU, making processing much faster.
- **Storage:** An SSD is mandatory. Traditional HDDs are too slow and cause major delays during launch and database updates.

Network & Accessories

- **Cameras & Connection:** Connect IP cameras using wired Ethernet instead of Wi-Fi to prevent video lag or missing frames.
- **Attendance Hardware:** Use a USB webcam or a secondary IP camera pointed at the staff entry register.

Internet Access: Requires outbound access to the Central API Gateway to send lightweight data payloads.